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Collaborative Disassembly for Multiple Users in Virtual Reality

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Abstract—Disassembly Sequence Planning (DSP) is critical in product lifecycle management but computationally demanding, with traditional and machine learning approaches facing challenges in scalability and lacking the real-time responsiveness required for interactive collaboration in virtual reality (VR). We present Dynamic Parallel Disassembly and Tasking (DPDT), a novel framework tailored for immersive collaboration. DPDT integrates a spatial-temporal feature-based heuristic to accelerate search prioritization and employs a dynamic reactivation strategy to resolve inter-part dependencies during parallel execution. Furthermore, a proximity-aware task assignment algorithm optimizes the translation of these plans into efficient multi-user instructions. We conduct extensive evaluations on a large-scale dataset against multiple baselines, showing that DPDT significantly outperforms state-of-the-art baselines. Specifically, for complex assemblies, our method achieves a computational speedup of up to 4.7 times while improving planning reliability by approximately 9.3 percentage points. These algorithmic gains translate directly into operational efficiency: a 24-participant user study in VR confirms that DPDT significantly reduces task completion time, physical-effort proxies, and NASA-TLX workload, validating its potential for effective collaborative maintenance training.

Index Terms—Virtual Reality, Disassembly Sequence Planning, Collaborative Disassembly, Human-Computer Interaction.

1 Introduction

Disassembly Sequence Planning (DSP) is pivotal in various stages of a product's life cycle, including manufacturing, maintenance, repair, and recycling [1]. The inherent combinatorial complexity, where the number of possible sequences can explode exponentially with the number of components, makes these planning tasks exceptionally challenging [2].

Historically, addressing these challenges has involved various approaches, each with its own set of limitations. Searching and optimization based algorithms [3], [4], [5], while capable of finding optimal or near-optimal solutions, often grapple with extremely long computation times, particularly for complex assemblies. Moreover, these methods are frequently designed or tuned for specific object geometries or types, hindering their ability to be rapidly adapted or scaled to new, unseen models. This lack of generalization capability and speed poses a significant bottleneck in dynamic or rapidly changing operational environments. More recently, machine learning techniques have been applied to ASP/DSP, offering the potential to learn from data [6], [7], [8]. However, learned predictions can still include physically invalid disassembly steps, especially for unseen geometries. Consequently, to ensure a physically valid and correct disassembly sequence, multiple iterations of physical simulation are typically required for verification. If a predicted step fails in simulation, the system must backtrack and retry, leading to a cumulative increase in overall planning time and computational overhead. Moreover, existing DSP planners rarely expose multi-user collaborative execution as a first-class objective. For learning-guided and heuristic planners, predicted steps still require computationally expensive physical simulation checks, which can hinder the interactive responsiveness expected in VR disassembly applications.

This leaves a practical gap: existing DSP planners rarely combine physics-aware sequence planning with multi-user task allocation, workload distribution, and responsive VR guidance. DPDT addresses this gap by unifying algorithmic planning and user-specific task assignment in a cohesive pipeline for interactive collaborative VR. The present study is scoped to simulator-validated, predominantly translational removals of rigid parts without explicit tool constraints or complex rotational actions; it is therefore a simulation-based proof of concept for VR training and maintenance rehearsal, rather than evidence of immediate industrial readiness.

To address these challenges, we propose the Dynamic Parallel Disassembly and Tasking (DPDT) framework, a comprehensive pipeline designed to enable efficient multi-user collaborative disassembly in virtual environments. First, we develop a spatial-temporal feature-based disassembly priority prediction mechanism. By synthesizing geometric features, such as visibility and center distance, with dynamic temporal data from attempt history, this module effectively ranks candidate parts to accelerate the search for valid sequences. Second, we introduce a dynamic reactivation-driven parallel disassembly planning algorithm that identifies sets of parts removable concurrently and handles emergent inter-part dependencies by updating candidate queues online during planning. Finally, to bridge the gap between algorithmic planning and human collaboration, we propose a proximity-aware task assignment strategy that transforms the parallel plan into interactive guidance, optimizing task distribution among multiple users by minimizing virtual travel distance. Extensive experiments demonstrate that DPDT achieves significant improvements in both computational speed and planning reliability over state-of-the-art methods. Furthermore, a user study confirms that our framework

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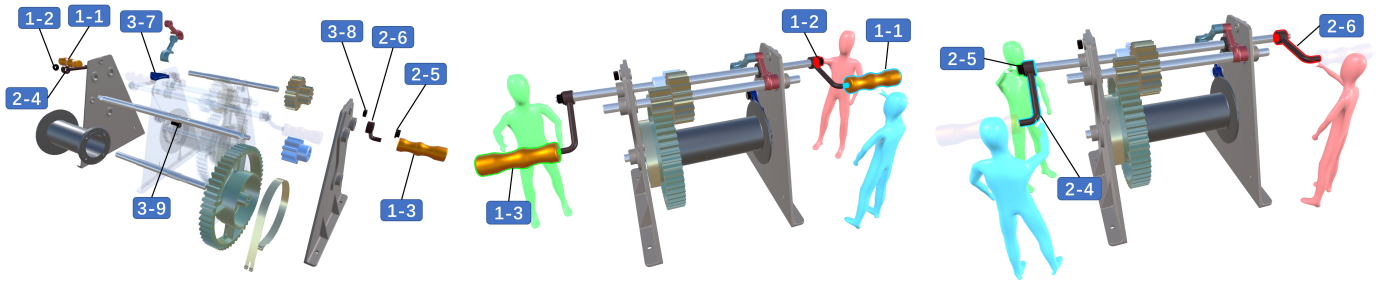


Fig. 1: The workflow of our Dynamic Parallel Disassembly and Tasking (DPDT) framework, illustrating the process from planning to guided collaborative execution on a winch assembly. **(a)** The complete parallel disassembly sequence generated by our planning algorithm, where each part is labeled with its `round-id` to indicate its removal order. **(b)** The first round of disassembly in virtual reality: the system highlights and assigns the initial set of removable parts (1-1, 1-2, and 1-3) to users based on proximity to minimize physical effort. **(c)** A subsequent round of the task: once the parts from the previous round are removed, the system dynamically guides users to the next set of available parts (2-4, 2-5, and 2-6), ensuring a continuous and efficient collaborative workflow.

substantially reduces task completion time and physical workload in collaborative VR scenarios. Fig. 1 shows the workflow of our dynamic parallel disassembly and tasking method.

In summary, our contributions are as follows:

- We present DPDT, an integrated task-level framework for collaborative VR disassembly that connects physics-aware sequence planning, multi-user task assignment, and interactive guidance in a shared virtual environment.
- We develop an adaptive disassembly-priority strategy that combines spatial cues with failed-attempt feedback, allowing the planner to update candidate ordering during search rather than relying on a fixed heuristic or a precomputed dependency graph.
- We introduce a dynamic reactivation mechanism for parallel disassembly planning. It uses online contact feedback from physics validation to handle dependencies that emerge during batched removal attempts, enabling more robust parallel planning without assuming a complete dependency graph in advance.
- We design a proximity-aware tasking and guidance layer that translates validated parallel plans into user-specific VR instructions, so collaborative execution considers both disassembly feasibility and user movement effort.
- We validate DPDT through large-scale planning experiments, ablation studies, and a 24-participant VR user study with NASA-TLX workload assessment, covering both algorithmic performance and user experience.

The remainder of this paper is organized as follows: Section 2 reviews related work in disassembly sequence planning, machine learning approaches, path planning, parallel disassembly, and collaborative task assignment. Section 3 details our proposed DPDT framework, including its part prioritization, planning, and task assignment algorithms. Section 4 describes the experimental setup, metrics, results, and ablation studies for the planning component of DPDT. Section 5 presents the user study conducted to evaluate the complete DPDT framework in a VR environment. Finally, Section 6 concludes the paper, outlines limitations and avenues for future research.

2 Related Work

2.1 Searching and Optimization Methods

Assembly Sequence Planning (ASP) and its inverse, Disassembly Sequence Planning (DSP), are fundamental problems in product lifecycle management [1]. Early research relied on graph-based representations, such as AND/OR graphs [9], to model feasible sequences. However, constructing these graphs typically requires predefined geometric and precedence constraints derived from CAD models or domain experts [10], [11]. Once feasible sequences are identified, optimization techniques including genetic algorithms and heuristic search are employed to find optimal sequences based on criteria such as assembly time, cost, or stability [12], [4], [13]. To manage the inherent complexity, the "assembly-by-disassembly" strategy is widely adopted [14], leveraging the more constrained nature of disassembly movements to reduce the search space. For rigid assemblies, a direct bijection exists between disassembly and assembly sequences, allowing an assembly plan to be derived by reversing a valid disassembly sequence [3]. Our DPDT framework replaces computationally intensive global optimization with a dynamic, heuristic-driven strategy guided by spatial-temporal prioritization, ensuring efficiency for interactive collaborative environments.

2.2 Machine Learning Methods

Recent advances in Machine Learning (ML) have opened new avenues for tackling ASP/DSP [6]. Reinforcement Learning (RL) agents learn (dis)assembly policies through simulated interaction [15], [16]. Graph Neural Networks (GNNs) predict part removability from graph representations encoding geometry and connectivity [7]. Transformer architectures model long-range dependencies in graph-structured sequence planning [17]. ASAP [8], included as our recent learning-guided baseline, combines physics-verified planning with GNN and heuristic ordering. While these methods can discover non-obvious strategies, they require large training datasets, struggle with out-of-distribution geometries, and their "black box" nature can be prohibitive in industrial contexts. In contrast, our heuristic-based approach is data-free, computationally lightweight, and fully interpretable.

2.3 Disassembly Path Planning and Physical Simulation Methods

Beyond determining the sequence, finding a collision-free path for each step is crucial. While sampling-based planners like PRM [20] and RRT [21] are prevalent, they struggle in the highly constrained configuration spaces typical of (dis)assembly [24]. Physics-based motion planning [25] directly simulates physical forces and interactions, making it advantageous for disassembly

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TABLE 1: Representative comparison of existing method categories and DPDT.

Method Category	Sequence	Physics	Parallel	Multi-User	Interactive VR
Search-based DSP [9], [14]	Sequential	No	No	No	No
Metaheuristic DSP [18], [19]	Sequential/Parallel	No	Async	No	No
Geometric path planning [20], [21]	Sequential	No	No	No	No
ML-based DSP [7], [17]	Sequential/Parallel	Sometimes	Limited	No	No
Physics-based DSP [5], [8]	Sequential/Parallel	Yes	Limited	No	No
XR-assisted systems [22], [23]	Guidance only	No	No	Limited	Yes
DPDT (Ours)	**Sequential/Parallel**	**Yes**	**Dynamic**	**Yes**	**Yes**

Note. "Async" denotes offline or population-level parallel search rather than synchronized parallel execution. "Sometimes" means that physics cues are used only in some variants or training settings. "Limited" indicates partial support that is not integrated with dynamic multi-user VR tasking. "Dynamic" denotes online queue updates from dependencies discovered during planning.

where intricate contact states and non-linear escape trajectories are common. Assemble Them All (ATA) [5] is the established physics-based reference for our work: it uses DiffRedMax [26], [27] with SDF-based contact resolution and provides the normalized assembly benchmark used in our experiments. ASAP [8] further builds on this line, making ATA and ASAP the two most relevant baselines for evaluating DPDT's planning core. We selected DiffRedMax over geometry-only collision checking or a different general-purpose rigid-body simulator because its SDF-based penalty contact model and implicit integration have already been used for contact-rich assembly and disassembly in ATA and ASAP. Reusing the same engine and thresholds also permits a controlled baseline comparison. Within DPDT, DiffRedMax is a simulated feasibility validator; a successful simulation does not by itself establish physical-world executability.

2.4 Parallel Disassembly Sequence Planning Methods

Parallel Disassembly Sequence Planning (PDSP) accelerates the process by identifying sets of components removable concurrently without interference. Research has explored metaheuristic optimization, such as Ant Colony [18] and Particle Swarm [19], to find efficient asynchronous parallel plans. Our parallel DSP algorithm (Algorithm 2) targets multi-agent collaboration with a dynamic reactivation strategy that handles dependencies emerging during batched disassembly attempts.

2.5 XR-Assisted Assembly and Disassembly Systems

Extended Reality (XR) technologies have increasingly been applied to industrial assembly and disassembly, providing immersive visualization, spatial guidance, and collaborative capabilities [22], [28]. Eswaran and Bahubalendruni [29] proposed an AR/VR-assisted autonomous disassembly framework, while their follow-up review [23] surveyed XR applications across the industrial assembly lifecycle. However, these systems primarily focus on guidance visualization — disassembly sequences are typically predefined or manually determined, without incorporating intelligent sequence planning. Our DPDT framework addresses this gap by coupling automated, physics-aware planning with immersive multi-user collaborative interaction in a unified pipeline. The representative comparison in Table 1 summarizes how these categories differ from DPDT along the dimensions most relevant to collaborative VR disassembly.

In summary, prior research has made important progress in sequence search, learning-based prediction, physics validation, parallel planning, and XR guidance. However, few existing systems jointly provide physics-aware sequence validation, dynamic handling of dependencies that appear during batched planning, task allocation for multiple human users, and interactive VR guidance evaluated with user performance and workload measures. As reflected in Table 1, DPDT is positioned not as a sequence planner only, a physics validator only, or a visualization interface only, but as an integrated task-level framework for collaborative VR disassembly.

3 METHOD

As illustrated in Fig. 2, DPDT connects planning and VR execution through three tightly coupled steps. First, spatial-temporal feature-based disassembly priority prediction (Section 3.1) synthesizes geometric and historical information to rank candidate parts before expensive simulation checks. Second, dynamic reactivation-driven parallel disassembly planning (Section 3.2) groups removable components into parallel rounds while updating inter-part dependencies from contact feedback. Third, proximity-aware task assignment (Section 3.3) translates the resulting plan into individualized VR guidance that reduces user movement and idle time.

3.1 Spatial-Temporal Feature-Based Disassembly Priority Prediction

The efficacy of any disassembly planner hinges on its ability to intelligently identify and prioritize viable candidate parts. DPDT therefore uses a spatial-temporal aware prioritization method that combines current geometric accessibility with feedback from previous failed attempts. This hybrid measure keeps the search focused on parts that are likely to be removable while adapting when physics validation contradicts the geometric estimate.

3.1.1 Spatial Features

Spatial features quantify the geometric state of a part in relation to the rest of the assembly at a single point in time. We use three signals that can be updated during planning without constructing an explicit precedence graph.

- **Visibility (VAR, MVAR, SVAR).** Visibility is estimated from six orthogonal depth-map views $(+x, -x, +y, -y, +z, -z)$. For each view, the Visibility Area Ratio (VAR) measures the visible projected area of the part relative to its total projected area. The maximum

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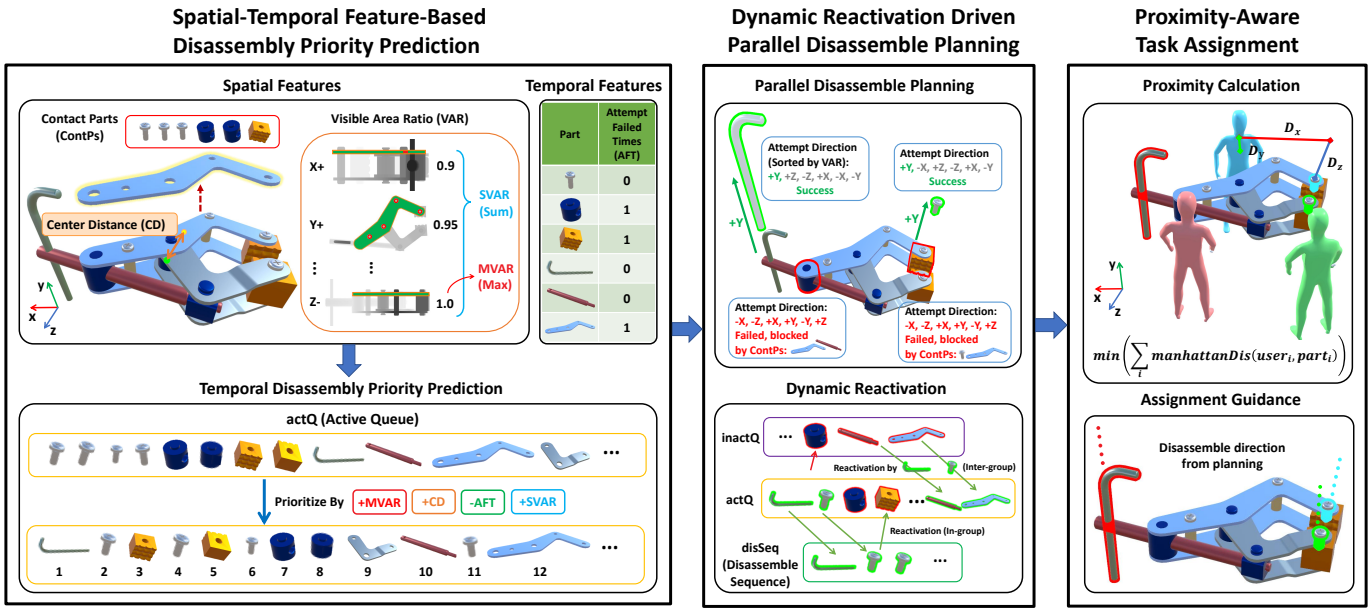


Fig. 2: The overall pipeline of our Dynamic Parallel Disassembly and Tasking (DPDT) framework. The process commences with Spatial and Temporal Disassembly Priority Prediction, where the system analyzes the assembly to compute spatial features, such as the Visibility Area Ratio and Center Distance, along with temporal data, including the number of Attempt Failed Times for each part. These metrics are synthesized to rank candidates in the active queue $actQ$, ensuring the most promising parts are addressed first. This prioritized queue then drives the Dynamic Reactivation Driven Parallel Disassembly Planning stage. Here, the algorithm attempts to remove multiple components concurrently and features a novel dynamic reactivation mechanism that intelligently resolves inter-part dependencies by promoting previously blocked parts from the inactive queue $inactQ$ back to the $actQ$ as obstructions are cleared. Finally, in the Proximity-Aware Task Assignment module, the successfully planned disassembly steps are distributed as tasks to users. This assignment optimizes the collaborative workflow by minimizing the total Manhattan distance between users and their assigned parts, and provides clear visual instructions through part highlighting and rendered disassembly paths to guide users effectively in the virtual environment.

over six views ($MVAR$) captures the best potential removal direction, while the sum ($SVAR$) measures global exposedness.

- **Center Distance (CD).** CD is the Euclidean distance from the part center to the assembly centroid. Peripheral parts usually have fewer neighbors and lower collision risk, so this feature supports a coarse outside-in ordering.
- **Contacting Parts ($contPs$).** When a physics attempt fails, the simulator reports the parts that blocked removal. This set is not a direct sorting feature, but it records dependency relations that static visibility cannot infer and is later used by dynamic reactivation.

These features reflect complementary geometric constraints. High visibility suggests a less occluded translational direction, large center distance suggests fewer surrounding constraints, and contact history reveals which constraints have actually appeared during physical validation. The feature set remains lightweight and interpretable. Instead of estimating a full dependency graph before planning, which can be expensive and brittle for complex assemblies, the planner uses visibility and periphery cues to choose promising candidates and lets the physics engine reveal the remaining contact constraints. The resulting $contPs$ records are then reused by later rounds, so the representation becomes more informative as the search proceeds.

3.1.2 Temporal Features

Temporal features introduce a memory-based, adaptive component to the planning process, allowing the algorithm to learn from its own search history and avoid redundant computations. Specifically, we utilize the number of failed attempts to track historical interactions. We maintain the Attempt Failed Times (AFT) counter for each part and increase it when `AtmptDis` fails. A high AFT acts as a penalty that lowers priority, preventing the planner from repeatedly trying parts that appear accessible in depth maps but remain physically constrained by tight tolerances or interlocking geometry. In this way, AFT bridges the gap between apparent removability from geometry and actual removability from simulation.

3.1.3 Spatial-Temporal Aware Prioritization

The prioritization routine takes the active queue $actQ$ and a persistent state table $STInfo$, which stores AFT and the latest $contPs$ for each part. It returns the same queue reordered by estimated removability, together with the updated state table. For each candidate part p and viewpoint v, the target-part depth map is rendered and compared with the corresponding whole-assembly depth map. The Visibility Area Ratio is computed as:

$$VAR_{p,v} = \frac{\sum \mathbb{I}(D_{p,v} = D_{total,v})}{\sum \mathbb{I}(D_{p,v} < \texttt{maxD})} \tag{1}$$

where $D_{p,v}$ is the depth map of part p, $D_{total,v}$ is the depth map of the entire assembly, and $\mathbb{I}$ is the indicator function. Pixels with depth values below the background threshold `maxD` define the projected part area. From the six VAR values we compute $MVAR$ and $SVAR$, and in parallel compute CD from the part's bounding-box center to the assembly centroid. Whole-assembly depth maps

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are cached within each STAwPri call using lightweight off-screen rasterization, and the reported planning time includes this visibility overhead.

The active queue is then sorted lexicographically by $(-MVAR, -CD, AFT, -SVAR)$, where negative signs denote descending order. This ordering is an empirical greedy design rather than an optimality guarantee: it first tests the cue most directly associated with the existence of an open translational direction, then uses peripheral position, search history, and global exposedness to resolve progressively less direct cases. The key is applied as follows:

1) $-MVAR$**.** Parts with the highest maximum visibility are considered first, because a strong directional view is the most direct proxy for an unobstructed translational path.
2) $-CD$**.** Among parts with similar visibility, components farther from the assembly center are preferred, following an outside-in disassembly intuition.
3) AFT**.** Parts with fewer failed attempts are prioritized so that geometrically exposed but physically constrained candidates do not dominate repeated checks.
4) $-SVAR$**.** Global exposure is used as the final tie-breaker when the previous cues are similar.

By integrating geometric computation with this adaptive sorting strategy, STAwPri places the most promising candidates at the front of $actQ$ and reduces unnecessary feasibility checks. The ordering follows the common geometry-then-validation pattern in physics-aware disassembly planning [3], [8], but adds search-history feedback to avoid wasting checks on candidates whose apparent accessibility has already proven misleading. We validate this design empirically through the ablation study in Section 4.4.

Algorithm 1: Spatial-Temporal Aware Prioritization(STAwPri)

Input: Active queue $actQ$, persistent state table $STInfo$
Output: Spatial-Temporal aware prioritized active queue $actQ$, updated $STInfo$
for *dir in* $[x, -x, y, -y, z, -z]$ **do**
  $DMPList[dir] \leftarrow$ `GetDM`$(dir, PList)$
for *part in* $actQ$ **do**
  for *dir in* $[x, -x, y, -y, z, -z]$ **do**
    $DMpart \leftarrow$ `GetDM`$(dir, part)$
    $STInfo[part].VAR[dir] \leftarrow$ `Sum`$(DMpart = DMPList[dir])$ / `Sum`$(DMpart < maxD)$
  $STInfo[part].MVAR \leftarrow$ `Max`$(STInfo[part].VAR)$
  $STInfo[part].SVAR \leftarrow$ `Sum`$(STInfo[part].VAR)$
  $STInfo[part].CD \leftarrow$ `Distance`$(part.center, PList.center)$
$actQ$.`Sort` **by** $-MVAR$, $-CD$, AFT, $-SVAR$ **in** $STInfo$
return $actQ$, $STInfo$

3.2 Dynamic Reactivation-Driven Parallel Disassembly Planning

To meet the demands of interaction-oriented, multi-user collaborative VR, the planning stage converts the priority queue into rounds of concurrently executable removals. Its core objective is to identify components that can be scheduled in the same disassembly round. Batched attempts, however, introduce a state-dependent challenge: removing one part in a round can change whether another candidate in the same round should be considered blocked or temporarily obstructed. The inputs are the model part list *PList* and the maximum parallelism number *PalNum*; the outputs are the parallel disassembly sequence *disSeq* and the suggested collaboration user count *userCnt* (Algorithm 2). Each iteration starts with the `TDEL` termination check, which bounds both computation time and search depth. The same limits are used for all methods in Section 4, so runtime comparisons are made under identical interaction-oriented constraints. The `TDEL` thresholds are implementation parameters rather than part-specific heuristics. In collaborative VR, planning may be performed before a session or during task preparation, and it should terminate predictably when an assembly contains invalid contacts or overly constrained parts. Under the same limits, runtime differences therefore reflect the quality of prioritization, batching, and reactivation rather than unconstrained search.

Within each round, the system first prepares the candidate parts. `STAwPri` sorts the active queue $actQ$, the top *PalNum* candidates are moved into the temporary parallel queue $PalQ$, and DiffRedMax [26] evaluates their removal attempts using a consistent assembly state and collision setting. The successful set *sucPs* is appended to the sequence with the current round ID, and *userCnt* is updated from the observed parallel throughput so that the downstream VR task assignment has a practical estimate of the useful collaborator count. After simulation, DPDT updates the queues from contact evidence rather than treating every failure equally. The update distinguishes two dependency cases.

- **Inter-group Reactivation.** If a successfully removed part appears in the stored *contPs* of an inactive part, that inactive part is returned to $actQ$ because one of its blockers has disappeared. This opens newly feasible branches for later rounds.
- **In-group Reactivation.** If a failed part collided with another part that succeeded in the same batch, the failure is considered temporary. The part remains active for the next round without increasing AFT, because the obstruction will be absent after the current batch is committed.

Failures that satisfy neither condition are treated as persistent constraints. The corresponding part receives an increased AFT and is moved to the lower-priority inactive queue $inactQ$, so it is not immediately reinserted into the active candidate set unless its blocking condition changes or the active queue later needs replenishment.

Consider a five-part case with $PalNum = 3$ and $PalQ = \{A, B, D\}$, while an inactive part C stores $contPs(C) = \{A\}$ from an earlier failed attempt. If DiffRedMax reports that A succeeds, B fails against A, and D fails against an unremoved part E, then committing A immediately reactivates C because its recorded blocker has disappeared. The failure of B is treated as temporary because its blocker is removed in the same batch, so B stays active without increasing AFT. In contrast, D is deferred to $inactQ$ and receives a larger AFT because its blocker remains. This example corresponds to the queue transitions in Fig. 2: DPDT updates dependencies online from simulation contacts rather than precomputing a fixed dependency graph. Unlike conventional reactive scheduling, which usually reacts to disturbances in a known task graph, DPDT discovers blocking relations while the plan is being

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generated, making the dependency representation incremental and simulation-driven.

At the planning-loop level, the main costs are the six-view visibility update inside STAwPri, STAwPri's $O(k \log k)$ sorting for k active candidates, and up to *PalNum* physics attempts per round. Physics cost scales with batch size and geometric complexity. Runtime remains bounded by TDEL, which was set to 3600s per model and 120s per part in our experiments. When removals succeed, the number of remaining parts decreases monotonically; when no feasible part is confirmed, candidates are deprioritized or deferred with accumulated AFT. Reactivation can revisit deferred parts, but the same time and depth limits prevent unbounded cycling. At the end of each round, a round-finalization step replenishes *actQ* from *inactQ* if the active part count falls below *PalNum*, then increments the round counter and continues until all parts are processed or TDEL is reached.

Algorithm 2: Dynamic Reactivation-Driven Parallel Disassembly Planning

Input: Model part list *PList*, parallelism number *PalNum*
Output: Model disassembly boolean status, disassembly sequence *disSeq*, suggested collaboration user count *userCnt*
actQ ← *PList*, *inactQ* ← `new Queue()`
disSeq ← `new Queue()`, *disRound* ← 0, *userCnt* ← 0
$STInfo \leftarrow \{part: (AFT = 0, contPs = \emptyset) \mid part \in PList\}$
while *PList*.`Count()` > 0 **do**
  if `TDEL()` = *True* **then**
    return *False*, *disSeq*, *userCnt*
  actQ, *STInfo* ← `STAwPri`(*actQ*, *STInfo*)
  `PalQ` ← *actQ*.`DeQ`(*PalNum*) *results*.`Clear()`
  for *part in PalQ* **do**
    results[*part*] ← `AtmptDis`(*PList*, *part*, *STInfo*[*part*].*VAR*, *STInfo*[*part*].*AFT*)
  sucPs ← *results*(*status* = *Success*)
  userCnt ← ⌊(*userCnt* ∗ *disRound* + *sucPs*.`Count()`)/(*disRound* + 1)⌋
  for *part, status, path, contPs* ***in*** *results* **do**
    if *status* = *Success* **then**
      PList.`Remove`(*part*)
      disSeq.`EnQ`(*part*, *path*, *disRound*) **for** *inactP* ***in*** *inactQ* **do**
        if *STInfo[inactP].contPs*.`Contain`*(part)* **then**
          inactQ.`Remove`(*inactP*)
          actQ.`EnQ`(*inactP*)
    else
      STInfo[*part*].*contPs* ← *contPs*
      if *contPs*.`Contain`*(any p* ***in*** *sucPs)* **then**
        actQ.`EnQ`(*part*)
        continue
      else
        STInfo[*part*].*AFT* += 1
        inactQ.`EnQ`(*part*)
  if *actQ*.`Count()` < *PalNum* **then**
    actQ.`EnQ`(*inactQ*.`DeQ`(*PalNum-actQ*.`Count()`))
  disRound += 1
return *True*, *disSeq*, *userCnt*

DiffRedMax [26] provides the feasibility validation in our planner, using the penalty-based contact model and implicit time integration adopted in prior disassembly work [5], [8]. Remaining sources of simulation error include residual part interpenetration after preprocessing, simplified contact and friction assumptions, the fixed gravity setting, and numerical integration error from discrete time stepping. These errors do not invalidate the queue logic, but they can create failed attempts that are not purely caused by true assembly dependencies. DPDT mitigates this by updating priorities and contact dependencies online: repeated failures lower priority through AFT, while reactivation prevents a temporary collision from being treated as a permanent blocker. Physical deployment would still require real-world validation to quantify the simulation-to-reality gap. Consequently, the current findings should be interpreted as a simulation-based proof of concept for VR training and maintenance rehearsal. Friction variation, compliance, manufacturing tolerances, tool access, and other deployment-specific constraints must be represented and validated on physical assemblies before a generated plan can be considered execution-ready.

3.3 Proximity-Aware Task Assignment

Once a parallel disassembly plan is generated, consisting of parts grouped into sequential rounds, DPDT distributes the currently available tasks to users in the collaborative VR environment. The goal is not only to execute the plan correctly, but also to minimize physical effort and maximize team efficiency. This stage therefore couples proximity calculation, task assignment, and visual guidance.

3.3.1 Proximity Calculation and Assignment

The assignment algorithm takes the planned sequence *disSeq* and user count *userCnt*, and returns task assignments *tskAssign* (Algorithm 3). At each update, it identifies available users, retrieves the tasks from the earliest unfinished disassembly round, computes the minimum-cost matching, updates user states, and removes assigned parts from the pending sequence. Although Algorithm 3 is written as a loop for clarity, the VR implementation triggers the same logic whenever a user completes a task or a new round becomes accessible. This event-driven behavior is important in collaborative VR because users rarely finish at exactly the same time; the system can immediately reassign the next valid task to an idle user instead of waiting for all collaborators to synchronize. The design preserves the precedence structure of the parallel plan: users can only receive parts from the minimum remaining round ID, while tasks within that round can be distributed according to proximity. The assignment layer therefore does not change the physics-validated disassembly order; it only decides which available user should perform each currently valid task. In this sense, DPDT separates what is physically valid from who should execute it, allowing the planner to remain conservative while the VR layer adapts to asynchronous human progress.

To quantify the cost of assigning a user to a task, we calculate the 3D Manhattan distance (L_1 norm) between the user's virtual position and the centroid of the target part. This distance acts as a conservative proxy for collaborative VR movement, where users navigate around a central assembly within shared workspace boundaries and adjust vertical position through discrete controller inputs. Compared with a straight-line Euclidean shortcut, the Manhattan cost penalizes diagonal assignments that would cut across

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the assembly or a constrained shared workspace. The user-study movement measures evaluate how this assignment cost translates into recorded physical effort.

The `AssignMinManhatDis` function solves for the set of pairings that minimizes the total sum of these distances. Formally, let U_{avail} be the set of available users and P_{next} be the set of candidate parts for the current round. We seek an optimal assignment mapping $\mathscr{M} : U' \to P'$, where $U' \subseteq U_{avail}, P' \subseteq P_{next}$ that satisfies:

$$\min_{\mathscr{M}} \sum_{(u,p)\in\mathscr{M}} (|x_u - x_p| + |y_u - y_p| + |z_u - z_p|) \tag{2}$$

where (x_u, y_u, z_u) and (x_p, y_p, z_p) are the user position and part centroid. This is a Linear Assignment Problem, solved with the Hungarian algorithm in $O(n^3)$ time for n users. In the practical range of collaborative VR sessions, even teams with around a dozen users lead to only small matching matrices, so this cost is negligible compared with physics-based planning. Moreover, the assignment is invoked only at task-allocation events, such as when a user becomes idle or a new round becomes accessible, rather than as a continuous per-frame optimization.

Algorithm 3: Proximity Calculation and Assignment

Input: Planned disassembly sequence *disSeq*, user count *userCnt*
Output: Task Assignments *tskAssign*
Initialize *userCnt* users as *userL*, *tskAssign* ← `new List()`
while *disSeq*.`Count()` > 0 **do**
  avilU ← *userL* where *user.task* is `null`
  if *avilU*.`Count()` = 0 **then**
    continue
  minR ← `Min`(*disP.disRound* in *disSeq*)
  avilP ← `new Queue()`
  for *disP in disSeq* **do**
    if *disP.disRound* = *minR* **then**
      avilP.`EnQ`(*disP*)
  if *avilP*.`Count()` = 0 **then**
    break
  tskAssignRound ← `AssignMinManhatDis` (*avilU*, *avilP*)
  for *(user, part)* ***in*** *tskAssignRound* **do**
    user.task ← *part*
    disSeq.`Remove`(*part*)
  tskAssign.`AddAll`(*tskAssignRound*)
return *tskAssign*

When *PalNum* exceeds the number of users, the available parallel tasks form a buffer from which the algorithm selects the lowest-cost subset. This is particularly useful in collaborative VR because users finish at different times; keeping multiple valid tasks available helps maintain workflow continuity even when the workforce is limited. The buffer also prevents the assignment policy from becoming a rigid precomputed schedule: if a closer user becomes available or if another collaborator is still completing a longer manipulation, the next assignment can be chosen from the same validated round without violating disassembly feasibility.

3.3.2 Assignment Guidance in VR

After matching users to parts, the system communicates the assignment through visual guidance (Fig. 3). Assigned parts are highlighted with user-specific colors so collaborators can immediately identify their own targets in a shared assembly, and optional dotted trajectories with ghosted final positions show the removal paths computed during simulation. The two cues serve different purposes: highlighting resolves task ownership, while path visualization reduces uncertainty about where and how the part should move after selection. Together, they translate the optimized plan into coordinated actions that collaborators can follow without manually negotiating task ownership or inferring the intended removal direction.

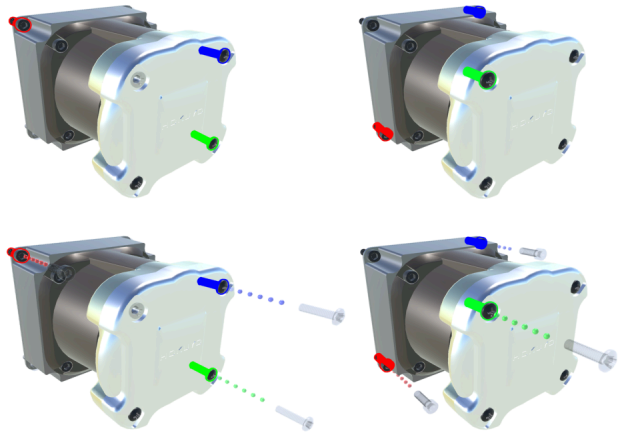


Fig. 3: Visual guidance workflow across two consecutive rounds. The upper images show user-specific target highlighting for active parts, and the lower images show the corresponding dotted removal trajectories and ghosted final positions used to guide users along the simulated disassembly paths.



Fig. 4: Multi-user collaborative disassembly in the VR environment. Three users interact with the same virtual assembly in a shared physical space, while individualized highlights and removal paths indicate each user's assigned parts.

4 Experiment

4.1 Experimental Setup

All planning experiments were conducted on a workstation with an Intel Core i7-13700K CPU, 64 GB of RAM, and an NVIDIA RTX 4090 GPU. Search, sorting, and physics simulation run on the CPU. DPDT additionally uses lightweight off-screen rasterization

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to obtain six-view depth maps, and the reported DPDT planning times include this visibility-rendering overhead. ATA and ASAP do not use these visibility maps, but all physics validation, BFS, TDEL, and hardware settings are kept identical across methods.

To evaluate the effectiveness of our DPDT framework, particularly its planning component, we conducted experiments using the large-scale dataset publicly released with the Assemble Them All (ATA) paper [5]. This dataset consists of numerous 3D models geometrically pre-processed to be watertight and normalized, and it supports direct comparison with recent ASAP planners [8] under the same physics-verified setting.

Our simulation environment aligns strictly with the baseline method ATA [5]. We utilize **DiffRedMax** [26], a differentiable simulator used in prior physics-based disassembly work with penalty-based contact modeling and implicit time integration. By adopting the same physics engine and collision detection thresholds as the baseline, we reduce environmental confounds and focus the evaluation on planning efficiency.

With BFS depth 15 and TDEL limits of 3600s per model and 120s per part, 3671 of 4196 local ATA assemblies passed method-independent mesh-loading and simulation-initialization checks. The *complex subset* was fixed a priori by part count, using the 382 retained models with > 20 parts, rather than by any method's success rate or runtime. These limits define the interaction-oriented operating range considered in this study. Each attempt uses 2000 simulation steps, gravity 9.8 m/s^2, six orthogonal visibility views along $\pm x$, $\pm y$, and $\pm z$, and parallelism $P_n \in \{1,2,3,4,5\}$. In the VR task-assignment conditions, proximity-aware assignment uses Hungarian matching with Manhattan distance.

4.2 Methods Compared and Metrics

We compared DPDT, whose planning core is Algorithm 2, with Assemble Them All (ATA) [5] and ASAP [8]. ATA is the established physics-based baseline, while ASAP provides recent GNN-based and volume-heuristic sequence planners. ASAP_GNN uses the released pretrained model, and ASAP_Heur uses its volume-based heuristic ordering. The ATA benchmark and the ASAP release are both based on Fusion360-style assembly data, supporting comparison under the same physics-verified setting. Both ASAP variants use the same DiffRedMax, BFS, TDEL, and P_n settings in the evaluated configurations.

All methods were evaluated with parallel disassembly numbers $P_n \in \{1,2,3,4,5\}$, which specify the maximum number of parts attempted in one planning round. The same P_n values and simulation limits are applied to every method, so the comparison isolates the effect of candidate prioritization and queue updating rather than different stopping rules. At $P_n = 1$, all methods operate sequentially; when $P_n > 1$, DPDT additionally uses contact-based reactivation to manage dependencies created by parallel attempts.

We used the following metrics for comparison:

- **Total Disassembly Plan Time** measures the time required to generate a full disassembly plan for a model, averaged across the corresponding test set.
- **Disassembly Step Success Rate**, abbreviated as SR, measures the proportion of predicted steps that pass physical simulation without collision or invalid states, averaged over all steps and models.

Methods with $P_n = 1$ were evaluated on all 3671 models. Configurations with $P_n \in \{2,3,4,5\}$ were evaluated on the 382-model complex subset, and $P_n = 1$ results on this subset are also reported for comparison. Higher-P_n tests focus on this fixed subset because high-part-count assemblies expose clearer parallel opportunities; all methods use the same subset.

4.3 Planning Results

The experimental results are presented and analyzed below, focusing first on the quality and reliability of the generated plans, and second on the computational efficiency.

4.3.1 Plan Quality and Reliability

Analysis of the Disassembly Step Success Rate (SR) demonstrates the superior reliability of our DPDT framework. DPDT consistently outperforms the baseline ATA across varying degrees of parallelism (P_n) and model complexities. The results are summarized in Table 2, and the trend with model complexity is shown in Fig. 5. Compared with the added ASAP baselines, DPDT also remains the best-performing method in every reported setting.

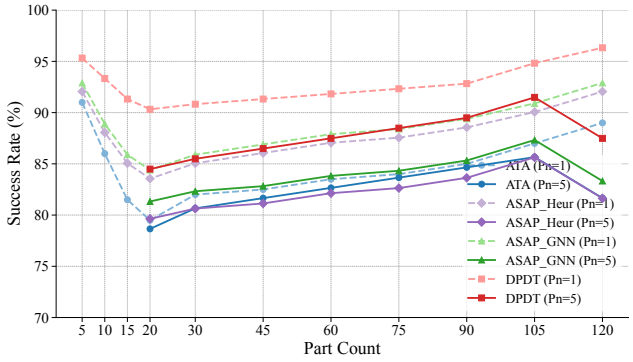


Fig. 5: Disassembly Step Success Rate (SR) with respect to assembly complexity.

In the sequential setting ($P_n = 1$), DPDT achieves an SR of 93.64% on the full dataset, improving over ATA's 86.90%. This margin widens to 9.28 percentage points on the complex subset with > 20 parts, validating the robustness of the spatial-temporal prioritization strategy in handling intricate geometries. As parallelism increases, the complexity of inter-part dependencies grows. While SR naturally declines under higher P_n values, DPDT exhibits greater resilience. At $P_n = 5$, DPDT maintains 87.47% SR, whereas ATA drops to 82.21%, ASAP_GNN reaches 83.65%, and ASAP_Heur reaches 81.96%.

The scalability analysis in Fig. 5 reinforces this finding, showing that DPDT consistently achieves higher reliability across the spectrum of part counts compared with the baselines. This demonstrates the robustness of DPDT and the effectiveness of its dynamic update strategy in handling the increased complexity and potential conflicts arising from parallel disassembly attempts. At $P_n = 5$, paired t-tests on per-model success fractions confirmed significant gains for DPDT over all three baselines. The comparisons with ATA, ASAP_GNN, and ASAP_Heur gave $t(381) = 8.28$, $t(381) = 5.52$, and $t(381) = 6.24$, respectively, all with $p < 0.001$. The corresponding Cohen's d values were 0.42, 0.28, and 0.32, and the 95% confidence intervals for the success-rate gains were $[4.02\%, 6.50\%]$, $[2.46\%, 5.18\%]$, and $[3.77\%, 7.25\%]$.

4.3.2 Planning Efficiency

In terms of computational efficiency, DPDT demonstrates a distinct advantage, particularly for complex assemblies where search

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TABLE 2: Disassembly Step Success Rate Comparison (%).

Method	3671 models	382 models (>20 parts)				
	$P_n = 1$	$P_n = 1$	$P_n = 2$	$P_n = 3$	$P_n = 4$	$P_n = 5$
ATA	86.90	84.42	83.78	83.25	82.65	82.21
ASAP_Heur	89.02	87.15	84.78	83.80	82.71	81.96
ASAP_GNN	89.86	87.99	85.07	84.37	84.10	83.65
DPDT	**93.64**	**93.70**	**89.91**	**88.55**	**87.92**	**87.47**

space explosion is a critical bottleneck. Table 3 and Fig. 6 summarize these findings.

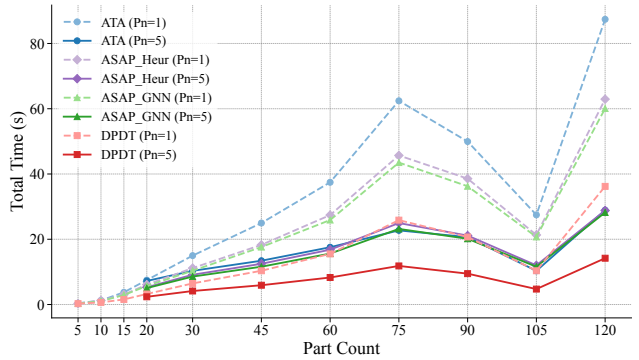


Fig. 6: Total Disassembly Plan Time with respect to assembly complexity. DPDT scales more efficiently than the baselines on high-part-count models.

For sequential planning ($P_n = 1$) on the complex subset, DPDT reduces the average planning time from 41.37s to 16.36s, achieving a 2.5× speedup over ATA and outperforming both ASAP_GNN (28.64s) and ASAP_Heur (31.28s). The efficiency gains become most pronounced under parallel configurations. With $P_n = 5$, DPDT reaches the lowest average time in Table 3 at 8.66s. This corresponds to a 4.7× speedup compared with ATA's sequential performance, remains 2.1× faster than ATA's best parallel result, and gives 2.0–2.1× speedups over the ASAP variants.

The time scalability analysis in Fig. 6 provides deeper insights. As expected, planning time generally increases with the number of parts. However, parallel DPDT, especially at $P_n = 5$, demonstrates superior time scalability and typically yields the fastest results for more complex models with higher part counts. The reported DPDT time includes the six-view visibility step, which took 10–30ms per planning round; this overhead is small relative to repeated physics validation. The trend is not strictly monotonic: at $P_n = 3$, some assemblies produce more in-group conflicts and reactivation checks than at $P_n = 2$, so the additional batch capacity is partly offset by queue churn before the larger buffers at $P_n = 4$ and $P_n = 5$ dominate.

Paired tests further support the timing results. For the sequential comparison on the complex subset, DPDT versus ATA gave $t(381) = -9.43$ with $p < 0.001$, Cohen's $d = -0.48$, and a 95% confidence interval of $[-30.22, -19.80]$s. At $P_n = 5$, paired tests on per-model planning times confirmed significant gains over ATA, ASAP_GNN, and ASAP_Heur. The three tests gave $t(381) = -9.13$, $t(381) = -7.48$, and $t(381) = -7.92$, all with $p < 0.001$; Cohen's d values were -0.47, -0.38, and -0.41; and the 95% confidence intervals were $[-11.70, -7.79]$s, $[-11.11, -6.50]$s, and $[-11.97, -7.21]$s. These results indicate that DPDT improves both the single-candidate search order and the high-parallelism scheduling behavior.

4.4 Ablation Studies

We conducted two ablation studies to isolate DPDT's main design choices.

4.4.1 *Contribution of Prioritization Features*

To evaluate the prioritization strategy, we tested variants that remove or reorder features from the sorting key $(-MVAR, -CD, AFT, -SVAR)$.

As shown in Table 4, the full MCAS ordering performs best in both the sequential setting with $P_n = 1$ and the parallel setting with $P_n = 5$. Reordering features slightly weakens performance: MCSA preserves a similar SR but is slower, while MSCA causes a clearer SR drop. Removing features produces larger degradation. CAS removes MVAR and therefore confirms directional visibility as the most informative cue. The progressive simplification from CAS to AS and then to S further shows that center-distance and attempt-history cues provide complementary information; SVAR alone is insufficient to guide the search reliably.

4.4.2 *Effectiveness of the Dynamic Reactivation Mechanism*

We also disabled the dynamic reactivation mechanism (DRM) on the complex subset at $P_n = 5$, where inter-part dependencies are most frequent. Compared with DPDT w/o DRM, full DPDT lowers planning time by 12.2% and improves SR by 1.36 percentage points. This indicates that reactivation avoids redundant checks and captures opportunities created within the same parallel step.

5 User Study

We conducted a user study comparing DPDT with ATA using Round-Robin assignment (ATA_RR), ATA using proximity-aware assignment (ATA_PA), and ASAP-GNN using Round-Robin assignment (ASAP_GNN_RR). The study evaluates task efficiency, movement effort, and subjective workload across different object complexities and collaborator counts.

5.1 Participants and Experimental Design

We recruited 24 participants from our university, including 14 male and 10 female participants aged 19–30 years. The mean age was 23.4 years with a standard deviation of 2.8. All had normal or corrected-to-normal vision and self-reported no history of VR-induced motion sickness. All participants provided informed consent before taking part in the study. The study used Pico headsets in a Unity 2022.3.5f1c1 collaborative environment running on an Intel Core i7-13700K with an NVIDIA RTX 4090. As shown in

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TABLE 3: Total Disassembly Plan Time Comparison (s).

Method	3671 models	382 models (>20 parts)				
	$P_n = 1$	$P_n = 1$	$P_n = 2$	$P_n = 3$	$P_n = 4$	$P_n = 5$
ATA	6.700	41.367	29.758	24.135	19.343	18.400
ASAP_Heur	5.079	31.284	28.116	25.307	21.862	18.247
ASAP_GNN	4.806	28.642	25.918	22.574	19.037	17.462
DPDT	**2.856**	**16.358**	**14.512**	**15.905**	**10.095**	**8.657**

TABLE 4: Ablation study results for prioritization features on the complex subset. In variant names, M, C, A, and S denote *MVAR*, *CD*, *AFT*, and *SVAR*, respectively, with the letter order indicating lexicographic priority.

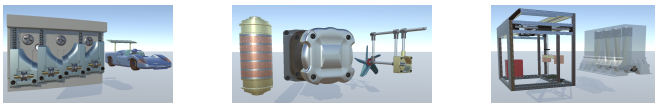
Variant	$P_n = 1$		$P_n = 5$	
	Time (s)	**SR (%)**	**Time (s)**	**SR (%)**
MCAS (full)	**16.358**	**93.70**	**8.657**	**87.47**
MCSA	17.626	93.69	8.981	87.12
MSCA	18.598	91.10	9.670	86.21
CAS (w/o MVAR)	23.172	90.30	10.612	84.50
AS (AFT+SVAR)	23.638	88.69	12.016	83.68
S (SVAR only)	23.514	88.88	11.806	83.80

TABLE 5: Ablation study results for the Dynamic Reactivation Mechanism (DRM) on the complex subset with $P_n = 5$.

Method Variant	**Total Time (s)**	**Success Rate (%)**
DPDT (full)	**8.657**	**87.47**
DPDT w/o DRM	9.863	86.11

Fig. 4, multiple users interacted with the same virtual assembly in a shared physical space while receiving individualized guidance.

Participants disassembled target objects according to the sequence generated by each method. They started from standardized virtual locations, and object placement was consistent across trials as shown in Fig. 7. Each assigned part was highlighted with a colored glow and path cue. Users navigated with the joystick, adjusted height with the X/A buttons for downward movement and the Y/B buttons for upward movement, and completed a step by touching the target part and pressing the trigger. Incorrect interactions were ignored.



(a) S1P1 (b) S2P2/S2P5 (c) S3P3

Fig. 7: Objects used in the user-study scenarios. S1P1 uses two objects with 60 parts, S2P2 and S2P5 use three objects with 66 parts, and S3P3 uses two objects with 126 parts.

We used a within-subjects design with four method conditions and four scenarios. The method conditions were:

- **DPDT.** Our full framework with spatial-temporal planning and proximity-aware task assignment.
- **ASAP_GNN_RR.** ASAP's GNN-based planner with Round-Robin assignment.
- **ATA_PA.** ATA's BFS-based plan with proximity-aware assignment, used to isolate the contribution of assignment.
- **ATA_RR.** ATA planning with fixed cyclic Round-Robin assignment, used as the full baseline because ATA does not define a multi-user allocation protocol.

The scenarios varied part count, parallelism, and user count:

- **S1P1.** 60 parts, $P_n = 1$, 1 user.
- **S2P2.** 66 parts, $P_n = 2$, 2 users.
- **S3P3.** 126 parts, $P_n = 3$, 3 users.
- **S2P5.** 66 parts, $P_n = 5$, 2 users.

Participants completed all scenario sessions. Collaborative trials were arranged as 12 two-person sessions for S2P2 and S2P5, and as 8 three-person sessions for S3P3. Each participant first completed a 5-minute VR practice trial with a tutorial object. Each scenario then included all four methods, counterbalanced with a Balanced Latin Square to reduce ordering effects. Participants completed the NASA-TLX questionnaire immediately after each method condition within every scenario, and mandatory breaks were enforced between method conditions to reduce fatigue and motion sickness. The logged measures were recorded separately for each participant, including assigned-part time, head movement, head rotation, and NASA-TLX ratings, so the primary analysis treats participant-level within-subject observations as user-experience measurements. Because waiting time is coupled within a team, we also inspected session-average summaries, with $N = 12$, $N = 12$, and $N = 8$ sessions for S2P2, S2P5, and S3P3, as a secondary sensitivity check; these summaries showed the same qualitative method ordering, while the participant-level analysis remains the primary evidence.

5.2 Metrics

We collected quantitative metrics to assess performance and user effort, along with a subjective workload assessment:

- **Time per Part**, abbreviated as T/Part and measured in seconds, is each participant's logged elapsed task time, including waiting between assigned parts, divided by the number of assigned parts.
- **Head Movement**, abbreviated as HeadM and measured in meters, is the total distance traveled by the user's head-mounted display.
- **Head Rotation**, abbreviated as HeadR and measured in degrees, is the total angular rotation of the user's head.
- **NASA-TLX** records subjective workload across Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration Level. For each participant and condition, we compute the raw overall score on a 0–100 scale as the mean of the six subscale ratings; the reported means and statistical tests then use these participant-level overall scores.

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Hand movement and hand rotation were also recorded as secondary logs and are not included in the main analysis.

5.3 User Study Results

We analyzed each metric with paired-samples t-tests between DPDT and every baseline in the same scenario, and then used two-way repeated-measures ANOVA to examine method, scenario, and interaction effects. Figure 8 gives the visual overview, while Table 6 reports the descriptive statistics, paired $t(23)$ values, and paired-samples effect sizes. Holm-Bonferroni correction was applied within each metric family of 12 planned pairwise comparisons. After correction, all Time per Part, Head Movement, and NASA-TLX comparisons remained significant at $p_{\text{adj}} < .05$. For Head Rotation, 10 of 12 comparisons remained significant, while the DPDT-versus-ASAP_GNN_RR contrasts in S1P1 and S2P5 were marginal after correction, both with $p_{\text{adj}} = .052$. Mauchly's tests for the Method effect were non-significant for all metrics with $p > .10$, so the reported Method statistics use the standard repeated-measures degrees of freedom. Scenario and Method-by-Scenario terms are reported with conventional degrees of freedom as omnibus context; Greenhouse-Geisser-corrected checks were considered where sphericity was violated, and the significance pattern was unchanged.

5.3.1 Task Efficiency

DPDT reduced Time per Part in all four scenarios, with the largest reduction in S2P2 and a clear gain even in the more complex S3P3 condition. The detailed means, paired tests, and effect sizes are listed in Table 6; all DPDT-versus-baseline time comparisons reached $p < .001$. The large effect sizes for Time per Part should be interpreted as end-to-end collaboration effects rather than pure locomotion effects, because this metric includes waiting between assigned parts. Round-Robin assignment can create substantial idle time when valid tasks are spatially or temporally imbalanced, so ATA_RR is expected to show larger differences than the decomposition baselines. The S2P5 result is also informative: allowing more planned candidates than active users gives the assignment algorithm a larger buffer, so idle time can be reduced without requiring five simultaneous operators.

The decomposition baselines help explain where the gain comes from. ASAP_GNN_RR improves over ATA_RR under the same Round-Robin assignment, which points to the value of stronger planning. ATA_PA improves the multi-user ATA conditions by changing only the assignment policy, but its gains are smaller than DPDT's. DPDT therefore benefits from both sides of the pipeline: it exposes better parallel task sets and then assigns those tasks to nearby users.

5.3.2 Physical Effort

DPDT also reduced the two physical-effort proxies in every scenario. The strongest movement reduction appears in S3P3, while the strongest rotation reduction appears in S2P2. The clearest effects appear in task time and head movement. Mean Head Rotation also decreased, although two contrasts against ASAP_GNN_RR became marginal after Holm correction. This pattern suggests that DPDT mainly improves sequencing, path availability, and allocation, whereas short-range viewing depends more on object-specific inspection needs. The baseline pattern is consistent with the intended separation between planning and assignment: better sequence planning reduces waiting and unnecessary target switching, while proximity-aware matching reduces the travel needed to reach the next valid part. ATA_PA confirms that assignment alone helps, but DPDT obtains larger reductions because it provides more suitable parallel task sets to assign.

The repeated-measures ANOVAs confirmed strong Method effects for Time per Part, Head Movement, and Head Rotation, with $F_{3,69} = 96.40$, 72.50, and 68.90, respectively, all at $p < 0.001$. The corresponding partial eta-squared values were 0.807, 0.759, and 0.750. Scenario effects were also significant for the three metrics, with $F_{3,69} = 68.70$, 61.30, and 58.20, again all at $p < 0.001$. Interactions were smaller but still significant, with $F_{9,207} = 5.84$, 4.67, and 4.35, respectively, all at $p < 0.001$. These effects indicate that DPDT's advantage remains consistent but varies in magnitude with task complexity and collaboration level.

5.3.3 Subjective Workload Assessment

NASA-TLX ratings show lower perceived workload with DPDT across all scenarios. Table 6 reports the overall workload scores and the paired statistics, and Table 7 summarizes the six NASA-TLX subscales. DPDT is lower on each subscale in every scenario, with the most visible margins usually appearing in Physical Demand, Effort, and Frustration. These subjective improvements are more conservative than the time and movement reductions because NASA-TLX also reflects individual confidence, collaboration dynamics, and scenario complexity.

The repeated-measures ANOVA on NASA-TLX overall scores showed a significant Method effect, $F_{3,69} = 38.60$, $p < 0.001$, $\eta_p^2 = 0.627$, and a significant Scenario effect, $F_{3,69} = 18.90$, $p < 0.001$, $\eta_p^2 = 0.451$. The interaction was smaller but still significant, $F_{9,207} = 2.10$, $p = 0.031$, $\eta_p^2 = 0.084$.

5.3.4 Scenario-Specific Observations

The improvement margin varied by scenario. S2P2 showed the largest time reduction, while S3P3 showed a smaller but still substantial reduction, suggesting that human operation speed becomes a stronger bottleneck for highly complex objects. The S2P5 condition further clarifies why P_n should be viewed as a planning buffer rather than simply as the number of users: even with two users, validating five candidate removals gives the assignment module more spatially diverse choices while preserving round-level precedence. The decomposition baselines support the same interpretation. ATA_PA improves over ATA_RR mainly by reducing unnecessary travel inside an already planned round, whereas ASAP_GNN_RR improves sequence quality but still assigns users cyclically. DPDT combines both effects by exposing better feasible choices and converting them into lower-effort execution.

The workload results are consistent with this decomposition but more conservative than the objective motion metrics. NASA-TLX reflects perceived effort, confidence, and frustration after a whole condition, so it is affected by collaboration complexity and individual tolerance as well as travel distance. The 24-participant study supports the reported user-experience trends, while standard HCI factors remain relevant. Counterbalancing reduces learning and fatigue effects without eliminating them completely, and future benchmarks should include longer maintenance sessions and VR-specific ergonomic factors such as headset comfort and controller fatigue.

6 Conclusion

We presented DPDT, a collaborative VR disassembly framework that combines spatial-temporal planning, dynamic reactivation

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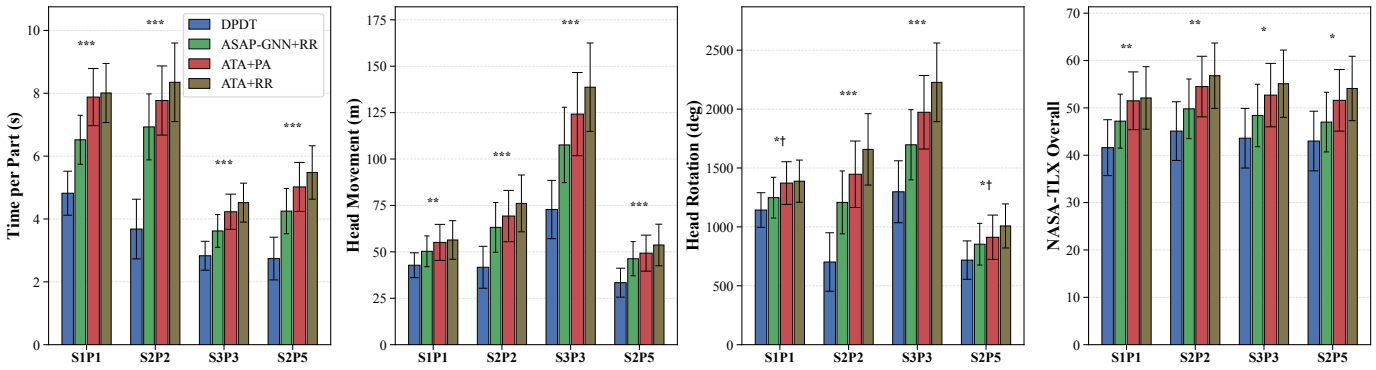


Fig. 8: User study results. Lower values indicate better performance, and error bars show standard deviation. Markers show the weakest uncorrected DPDT-versus-baseline category per scenario, with full pairwise tests and Holm notes in Table 6. One, two, and three asterisks denote uncorrected $p < .05$, $p < .01$, and $p < .001$; $*\dagger$ marks Head Rotation contrasts against ASAP_GNN_RR that became marginal after Holm correction.

TABLE 6: Integrated user-study results and DPDT-versus-baseline statistics. DPDT cells report Mean $\pm$ SD. Baseline cells report Mean $\pm$ SD followed by paired $t(23)$ and absolute paired-samples Cohen's d_z for DPDT minus that baseline. Negative t indicates a lower value for DPDT. Asterisks show uncorrected paired-test categories; $*\dagger$ marks uncorrected $p < .05$ Head Rotation comparisons that became marginal after Holm correction.

Metric	**Scenario**	**DPDT** M $\pm$ SD	**ASAP_GNN_RR** M $\pm$ SD; t / d_z	**ATA_PA** M $\pm$ SD; t / d_z	**ATA_RR** M $\pm$ SD; t / d_z
T/Part (s)	S1P1	4.82 ± 0.70	6.52 ± 0.78; -6.80 / 1.39^{***}	7.88 ± 0.91; -11.18 / 2.28^{***}	8.01 ± 0.94; -11.45 / 2.34^{***}
	S2P2	3.68 ± 0.95	6.93 ± 1.05; -9.65 / 1.97^{***}	7.77 ± 1.10; -11.80 / 2.41^{***}	8.35 ± 1.25; -12.37 / 2.52^{***}
	S3P3	2.83 ± 0.46	3.62 ± 0.52; -4.95 / 1.01^{***}	4.23 ± 0.56; -8.40 / 1.71^{***}	4.52 ± 0.62; -9.20 / 1.88^{***}
	S2P5	2.74 ± 0.68	4.25 ± 0.72; -6.45 / 1.32^{***}	5.02 ± 0.78; -9.45 / 1.93^{***}	5.48 ± 0.85; -10.74 / 2.19^{***}
HeadM (m)	S1P1	42.8 ± 6.7	50.3 ± 8.3; -3.02 / 0.62^{**}	55.1 ± 9.7; -5.05 / 1.03^{***}	56.4 ± 10.4; -5.30 / 1.08^{***}
	S2P2	41.7 ± 11.3	63.2 ± 13.4; -5.82 / 1.18^{***}	69.3 ± 13.8; -6.76 / 1.38^{***}	76.1 ± 15.3; -7.52 / 1.53^{***}
	S3P3	72.8 ± 15.7	107.6 ± 20.3; -6.28 / 1.28^{***}	124.2 ± 22.4; -7.38 / 1.50^{***}	138.7 ± 23.8; -8.18 / 1.67^{***}
	S2P5	33.4 ± 7.8	46.3 ± 9.2; -4.83 / 0.99^{***}	49.3 ± 9.7; -5.72 / 1.17^{***}	53.7 ± 11.2; -6.68 / 1.36^{***}
HeadR (°)	S1P1	1143 ± 147	1248 ± 173; -2.28 / $0.47^{*\dagger}$	1372 ± 181; -4.69 / 0.96^{***}	1388 ± 179; -4.91 / 1.00^{***}
	S2P2	702 ± 248	1208 ± 267; -6.18 / 1.26^{***}	1447 ± 282; -7.17 / 1.46^{***}	1658 ± 303; -8.83 / 1.80^{***}
	S3P3	1298 ± 263	1697 ± 298; -4.53 / 0.92^{***}	1973 ± 312; -6.87 / 1.40^{***}	2227 ± 334; -7.83 / 1.60^{***}
	S2P5	718 ± 163	853 ± 177; -2.38 / $0.49^{*\dagger}$	912 ± 188; -3.57 / 0.73^{**}	1008 ± 187; -5.13 / 1.05^{***}
NASA-TLX	S1P1	41.6 ± 5.9	47.2 ± 5.7; -3.18 / 0.65^{**}	51.5 ± 6.1; -5.76 / 1.18^{***}	52.1 ± 6.6; -6.10 / 1.25^{***}
	S2P2	45.1 ± 6.2	49.8 ± 6.3; -3.02 / 0.62^{**}	54.5 ± 6.4; -5.19 / 1.06^{***}	56.8 ± 6.9; -6.55 / 1.34^{***}
	S3P3	43.6 ± 6.3	48.4 ± 6.6; -2.52 / 0.51^{*}	52.7 ± 6.7; -5.00 / 1.02^{***}	55.1 ± 7.1; -6.35 / 1.30^{***}
	S2P5	43.0 ± 6.3	47.0 ± 6.3; -2.27 / 0.46^{*}	51.6 ± 6.5; -4.75 / 0.97^{***}	54.1 ± 6.8; -6.02 / 1.23^{***}

for parallel dependency management, and proximity-aware task assignment. STAwPri ranks candidates, dynamic reactivation turns physics feedback into queue updates, and proximity-aware assignment maps the validated parallel plan to users without changing the disassembly order. Experiments on 3671 models showed up to 4.7× speedup over ATA's sequential baseline and 5.3–9.3pp higher success rates than ATA across evaluated settings. In the 24-participant VR study, DPDT reduced time per part by 37.4–55.9%, head movement by 24.1–47.5%, and NASA-TLX workload by 20.2–20.9% compared with ATA_RR across scenarios. Comparisons with ASAP-GNN and ATA_PA further suggest that planning quality and assignment optimization contribute independently to the overall gain.

These results support DPDT as a task-level planning and guidance layer for VR training, maintenance rehearsal, and responsive operator guidance rather than a per-frame physical control system. The current prototype operates on preprocessed geometry and does not directly preserve CAD/PLM assembly hierarchies, mates, tolerances, material properties, or tool requirements. A concrete next step is a CAD/PLM connector that retains these semantics, translates them into simulator and planning constraints, and returns simulator-validated sequences to the product's digital thread for VR rehearsal; physical trials remain required before operational use. The benefit is strongest when assemblies expose multiple removable candidates: dynamic reactivation reveals feasible choices, and proximity-aware assignment selects convenient tasks from those choices. When precedence constraints are nearly linear, collaboration gains naturally diminish; future planners could detect such bottlenecks online and switch between parallel exploration and optimized sequential planning.

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TABLE 7: NASA-TLX subscale means for all user-study scenarios. Scores use a 0–100 scale, and lower values indicate lower perceived workload. Overall NASA-TLX statistics are reported in Table 6. MD denotes Mental Demand, PD Physical Demand, TD Temporal Demand, PF Performance, EF Effort, and FR Frustration. ASAP-RR denotes ASAP_GNN_RR.

Sc.	Method	MD	PD	TD	PF	EF	FR	Overall
S1P1	ATA-RR	53.2	64.1	54.3	40.8	51.9	48.5	52.1
	ATA-PA	52.7	63.8	54.2	40.3	51.1	46.9	51.5
	ASAP-RR	48.6	55.9	49.2	39.7	47.3	42.7	47.2
	DPDT	**44.2**	**46.3**	**43.7**	**38.2**	**43.8**	**33.4**	**41.6**
S2P2	ATA-RR	58.6	70.3	60.7	44.2	55.3	51.5	56.8
	ATA-PA	56.7	66.6	58.3	43.8	53.4	47.9	54.5
	ASAP-RR	51.7	59.2	54.1	42.3	49.6	42.0	49.8
	DPDT	**47.3**	**50.7**	**49.8**	**40.2**	**45.1**	**37.3**	**45.1**
S3P3	ATA-RR	57.3	69.2	59.3	43.3	54.2	47.1	55.1
	ATA-PA	55.1	65.3	56.7	42.8	52.3	44.0	52.7
	ASAP-RR	50.3	57.2	52.3	41.8	48.3	40.5	48.4
	DPDT	**46.8**	**48.7**	**48.2**	**39.3**	**44.3**	**34.1**	**43.6**
S2P5	ATA-RR	56.3	68.2	57.8	42.7	53.3	46.1	54.1
	ATA-PA	54.0	64.3	55.2	42.3	51.2	42.4	51.6
	ASAP-RR	49.2	56.1	51.0	41.3	47.2	37.3	47.0
	DPDT	**45.7**	**48.2**	**47.5**	**38.7**	**43.1**	**34.5**	**43.0**

Several limitations and future directions remain. The current DPDT pipeline assumes predominantly rigid parts and direct, tool-free translational removals validated by the simulator under simplified contact settings. It does not model full 6-DOF or helical extraction, threaded fasteners, tool geometry and approach direction, clearance, torque, ordered fastening actions, or deformable components such as cables and flexible harnesses. Future work will combine 6-DOF or hybrid sampling-based motion planning with explicit tool models and CAD/PLM-derived fastening constraints, then validate the resulting plans on physical maintenance tasks. Like other simulation-verified DSP pipelines, DPDT inherits errors from residual interpenetration, simplified contact and friction parameters, gravity assumptions, and numerical integration. The adaptive use of *AFT* and *contPs* reduces brittleness under failed attempts, but physical deployment still requires higher-fidelity product data and real-world validation. The ATA benchmark provides scale and comparability, but it is not a VR-specific industrial maintenance dataset with headset comfort, controller fatigue, communication, tool-use, and safety annotations. The evaluation is also bounded by TDEL and by assemblies practical for interaction-oriented VR guidance; much larger assemblies may require hierarchical decomposition or pre-session planning. Future work should construct or adopt such benchmarks, include expert technicians and longer sessions, and use assemblies from deployed CAD/PLM or digital-twin workflows. In these settings, deployment also needs explicit safety checks, operator override, role management, and collision-aware walking paths. Task assignment could become more user-aware by considering skill, fatigue, preferred working zones, communication cost, and controlled user agency.

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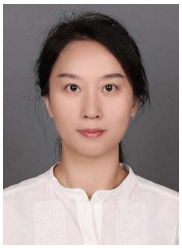
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